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DefenSYS: A Cross-Platform Capstone and PIT Management System with Secure Archiving and Curriculum Analytics

In the landscape of Information Technology education, Capstone and Project Integration Technology (PIT) courses stand as the definitive benchmarks of student proficiency, yet their management often relies on fragmented, manual processes that hinder efficiency and security. Faculty members frequently face logistical burdens during defenses, dependent on laptops or physical rubrics, while the lack of a centralized, secure repository leaves student intellectual property vulnerable to plagiarism and unauthorized reproduction. Furthermore, the absence of data-driven insights prevents administrators from utilizing project data to identify technology trends for curriculum improvement.

To address these critical gaps, this study proposes the **Development of a Cross-Platform Capstone and PIT Management System with Secure Archiving and Curriculum Analytics**. This integrated solution modernizes the academic ecosystem by offering a mobile-responsive evaluation tool for seamless grading, a secure read-only vault with dynamic watermarking to protect intellectual property, and an analytics engine to guide evidence-based curriculum planning.

Background of the Study

In the rapidly evolving landscape of Information Technology education, the Capstone and Project Integration Technology (PIT) courses stand as the ultimate benchmarks of student competency, yet many institutions remain hindered by fragmented, manual processes that compromise both efficiency and academic integrity. Current practices often burden faculty with cumbersome paper-based or laptop-dependent grading during defenses and leave student intellectual property vulnerable to plagiarism due to disorganized, insecure archiving. Furthermore, administrators lack data-driven mechanisms to analyze the content of these projects, often relying on observation rather than hard evidence to guide curriculum improvements. To address these critical gaps, this study proposes the **Development of a Cross-Platform Capstone and PIT Management System with Secure Archiving and Curriculum Analytics**, an integrated solution that streamlines evaluation through mobile-responsive tools, secures student outputs with dynamic watermarking and read-only access, and empowers the administration with predictive analytics to foster a relevant, future-proof IT curriculum.

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Statement of the Problem

Specifically, the current system presents the following gaps that this study aims to address:

1. Fragmented Workflow and Manual Evaluation

The department lacks a unified system for PIT and Capstone management, burdening administrators with manual team encoding. Furthermore, panelists are forced to rely on cumbersome physical rubrics or laptops during defenses due to the absence of a mobile-responsive grading tool.

2. Unstructured Archival and Poor Version Control

Current storage methods mix failed drafts with final manuscripts without a secure audit trail. Furthermore, the system lacks an automated mechanism to enforce document naming conventions, resulting in students uploading inconsistently named files. This absence of standardized file classification causes severe administrative confusion, hinders efficient retrieval, and risks the public circulation of unapproved or erroneous documents.

3. Vulnerability of Intellectual Property

Existing archives lack dynamic watermarking and read-only controls, leaving student manuscripts and source code exposed to unauthorized downloading and plagiarism.

4. Absence of Data-Driven Curriculum Planning

The department lacks automated analytics to track technology trends from student projects, preventing the administration from making evidence-based curriculum enhancements.

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Comparison of the related Problem

Features	Google Classroom / Moodle	GitHub / Bitbucket	Google Drive	Proposed System
Mobile-Responsive Grading	Yes (Generic)	No	No	Yes (Specialized)
Grade Calculation	Manual	No	No	Automated
Read-Only Access File	Partial	No (Clonable)	Partial (Static)	Yes (Dynamic)
Dynamic Watermarking (User ID)	No	No	No	Yes
Curriculum Trend Analytics	No	No	No	Yes
PIT & Capstone Integration	Manual	No	Partial	Integrated
Role-Based Access (Adviser vs Panel)	Limited	No	Limited	Highly Specific
Automated File Classification & Renaming (via ML)	No (Manual)	No	No	Yes

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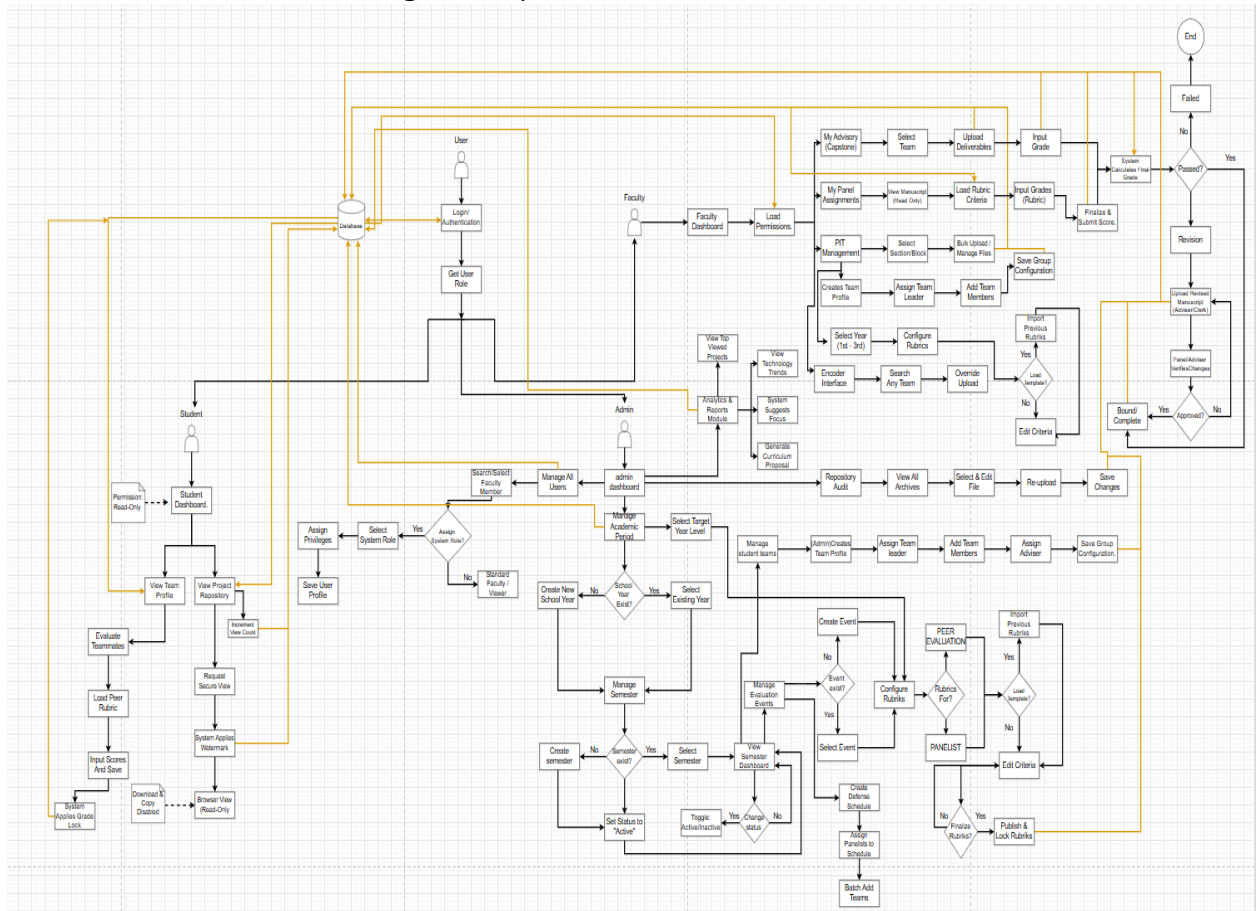
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Proposed Solution

This study proposes the development of the "Cross-Platform Capstone and PIT Management System," an integrated solution designed to modernize the evaluation and archiving workflow of the College of Information Technology. The system utilizes a robust hybrid architecture, combining a Web-Based Portal for administrative tasks and a Mobile Application for real-time faculty evaluation and student access. Furthermore, to ensure strict organizational compliance and eliminate manual sorting, the system integrates a Machine Learning-driven Natural Language Processing (NLP) pipeline utilizing TF-IDF and Naive Bayes algorithms. As illustrated in the system architecture, this AI component automatically extracts text from uploaded deliverables, classifies the document type, and enforces standardized filename conventions before securely committing the files to the digital vault.

Figure 1: System Architecture Overview



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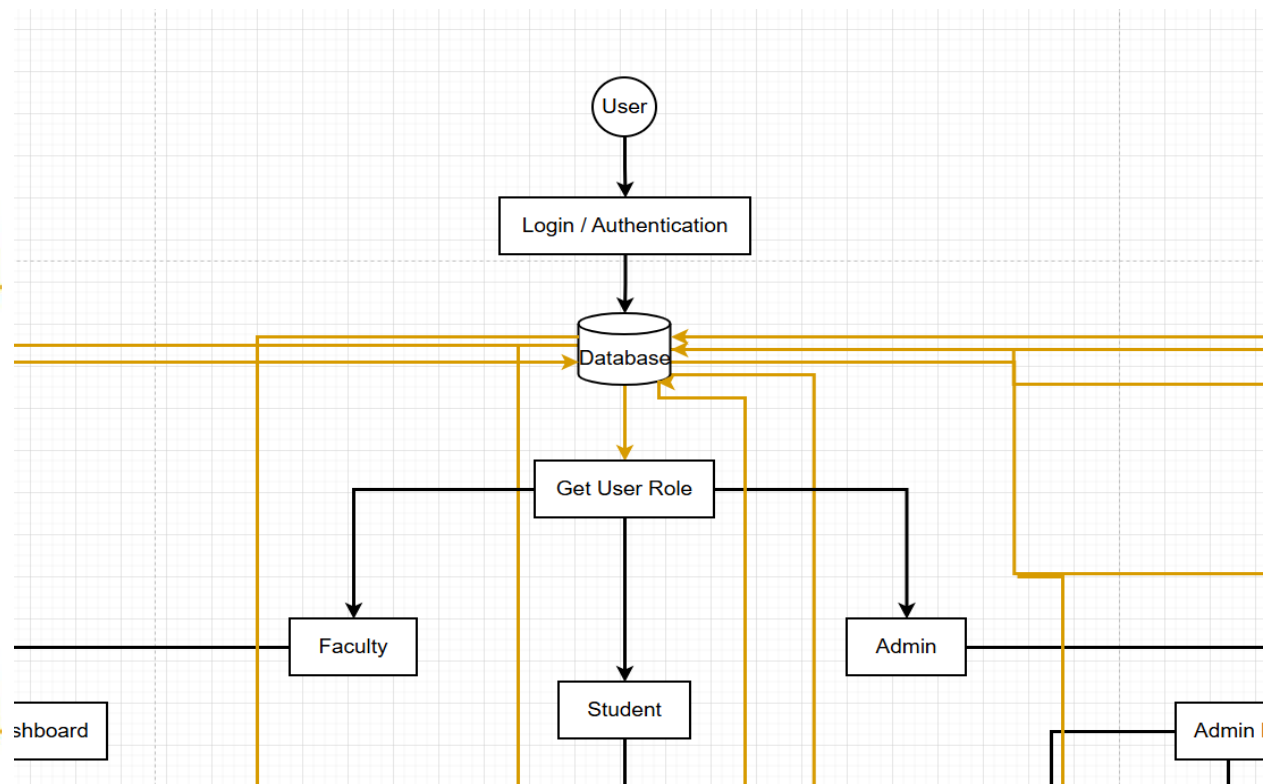
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As shown in the architecture above, the system is organized into four distinct modules:

But before we can access this four Modules we need to Login or Authenticate first

Figure 2: Login Process



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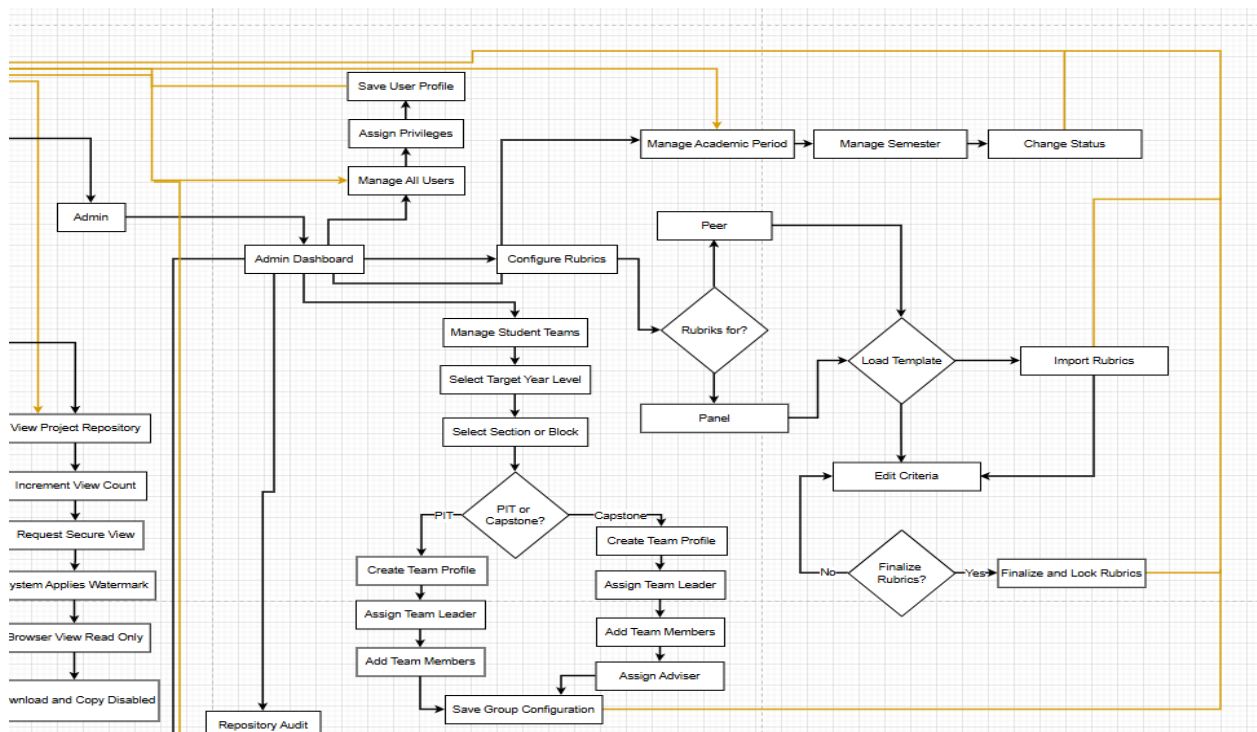
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1. The Administrative & Analytics Core (Web-Based)

- *Role:* Geared towards the Administrator and Clerk.
- *Function:* This module serves as the central control unit. It handles user management, academic year configuration, and rubric setup. Additionally, it features a 'Repository Audit Manager' that grants the Administrator global privileges to view all file versions (including failed or hidden submissions) and override file statuses. To ensure strict organizational compliance, this module integrates a Machine Learning-driven document classification pipeline. Upon file upload, a Natural Language Processing (NLP) model analyzes the extracted text to automatically identify the document type and enforces a standardized naming convention before securely archiving the file. This ensures complete control over the department's archival integrity and eliminates the administrative burden of manual file sorting. Crucially, it hosts the Decision Support System (DSS), which processes data from the repository to generate Curriculum Analytics, providing the department with actionable insights on technology trends.

Figure 3: Admin Side



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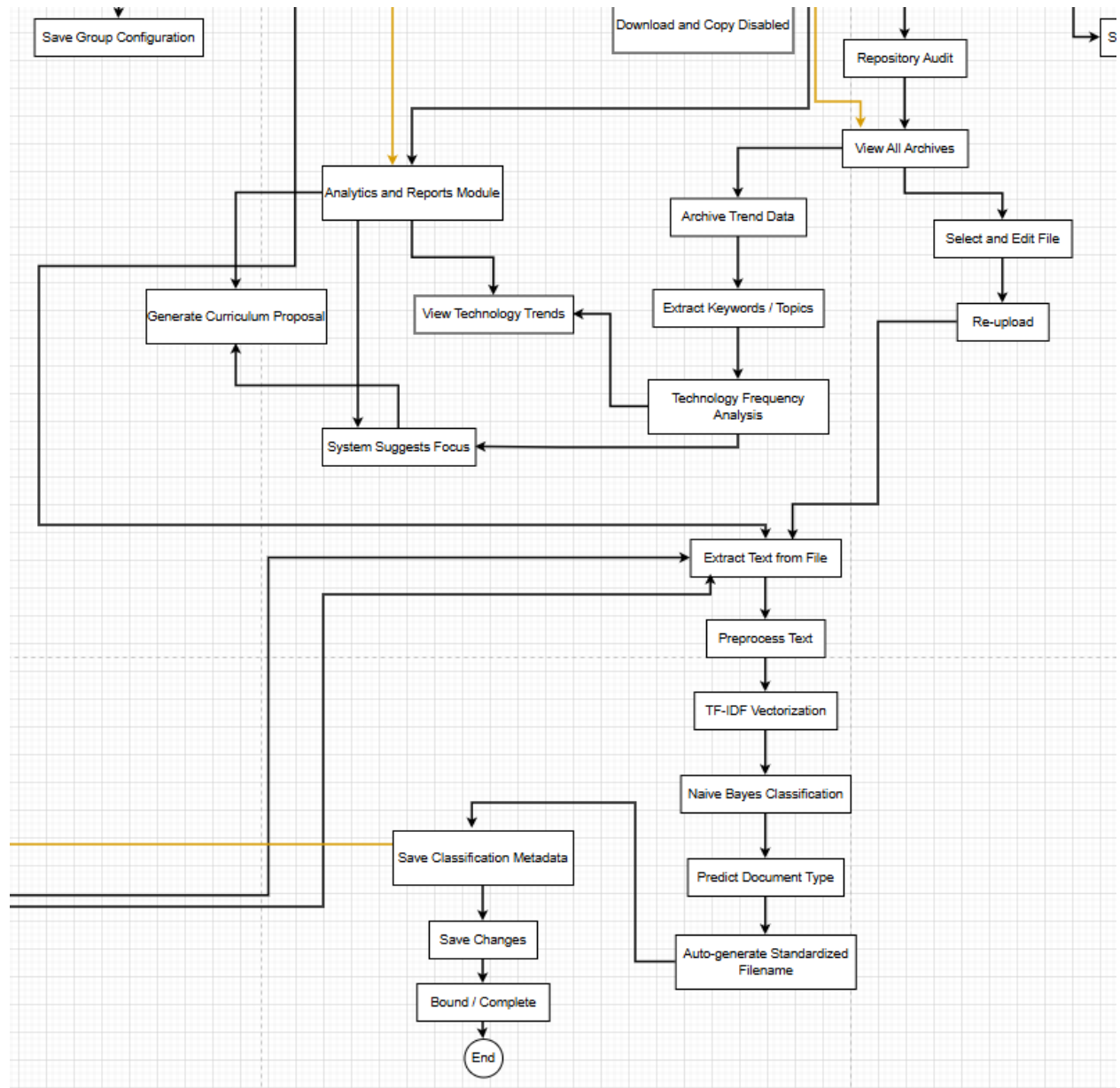
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Figure 4: Admin Side



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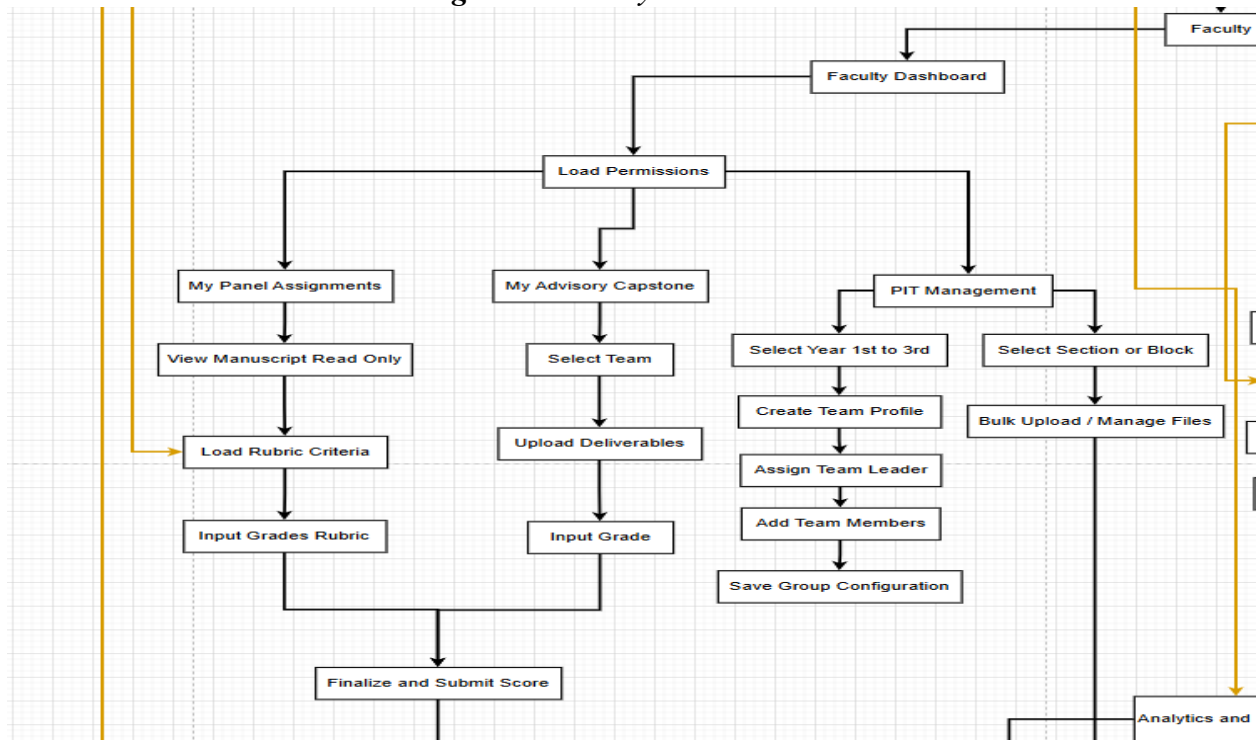
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2. The Evaluation & Grading Module (Hybrid Web/Mobile)

- *Role:* Designed for Advisers, Panelists, and PIT Leads.
- *Function:* This module facilitates a dynamic, multi-tiered evaluation system (e.g., a 50/30/20 grading split). It intelligently aggregates scores from Panelists, Advisers, and strictly isolated Peer-to-Peer evaluations to automatically compute the final systemic grade.
- **Mobile Interface:** Allows Panelists to input grades, view manuscripts, and approve revisions directly from their smartphones during on-site defenses.
- **Web/Mobile Interface:** Streamlines grading and manuscript viewing for faculty. Crucially, it empowers the PIT Faculty Leader to independently manage team formation for the entire lower-year cohort, decentralizing the process and significantly reducing the Administrator's manual workload.

Figure 5: Faculty Side



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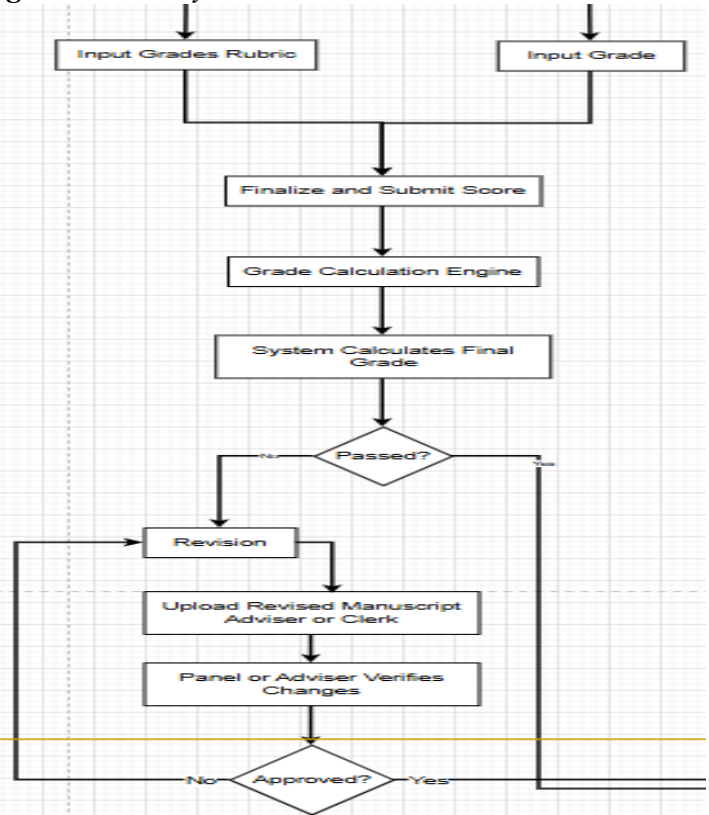
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Figure 6: Faculty Side



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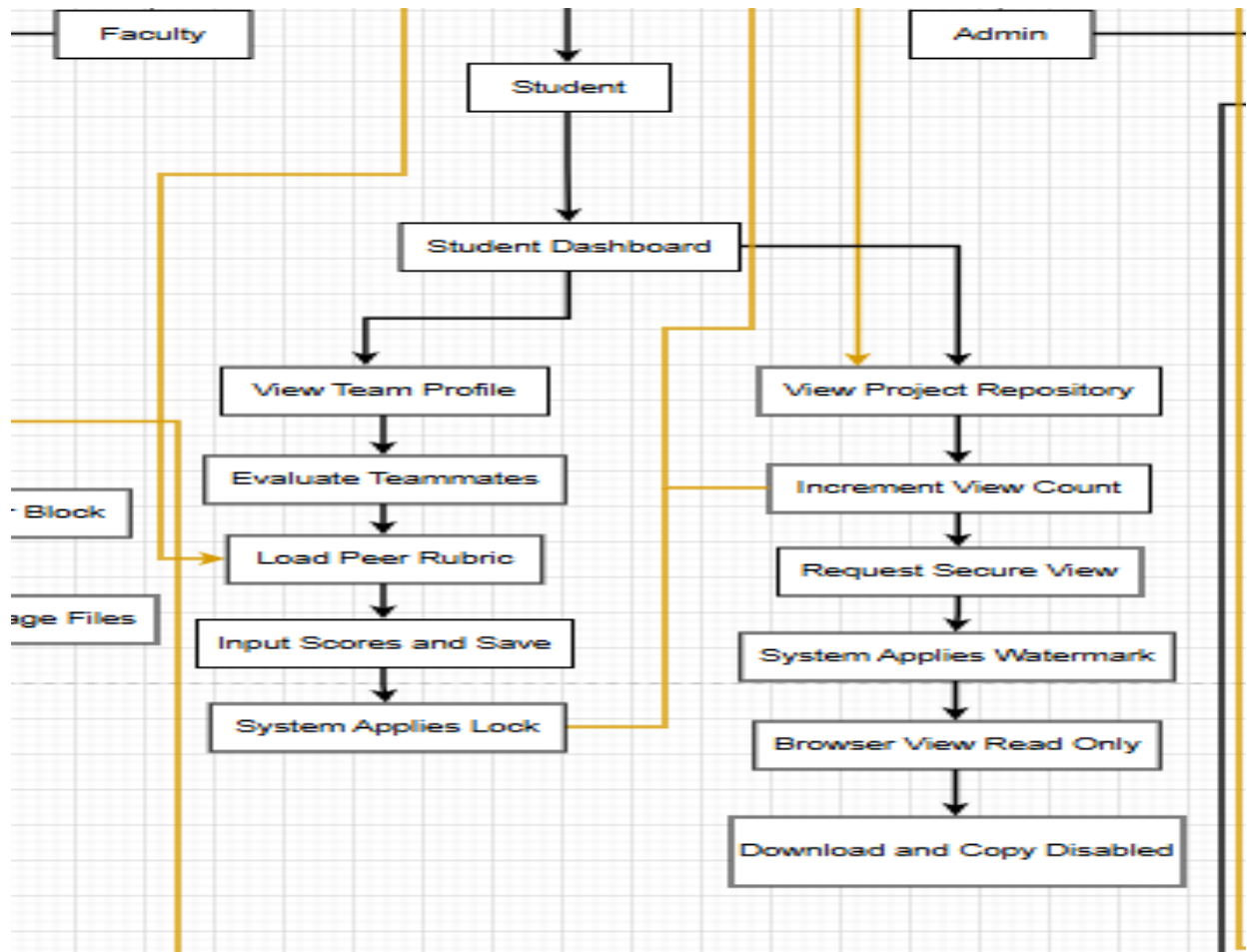
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3. The Secure Student Repository (Mobile/Web)

- *Role:* Accessible to Students.
- *Function:* While the repository remains strictly read-only, the Student Dashboard includes an isolated "Assigned Team" module. This allows students to view their official team profile and execute Peer-to-Peer evaluations using dynamic rubrics. To ensure absolute data integrity and prevent post-submission tampering, the system applies an automated "Grade Lock" the moment a student submits their peer evaluation.

Figure 7: Student Side



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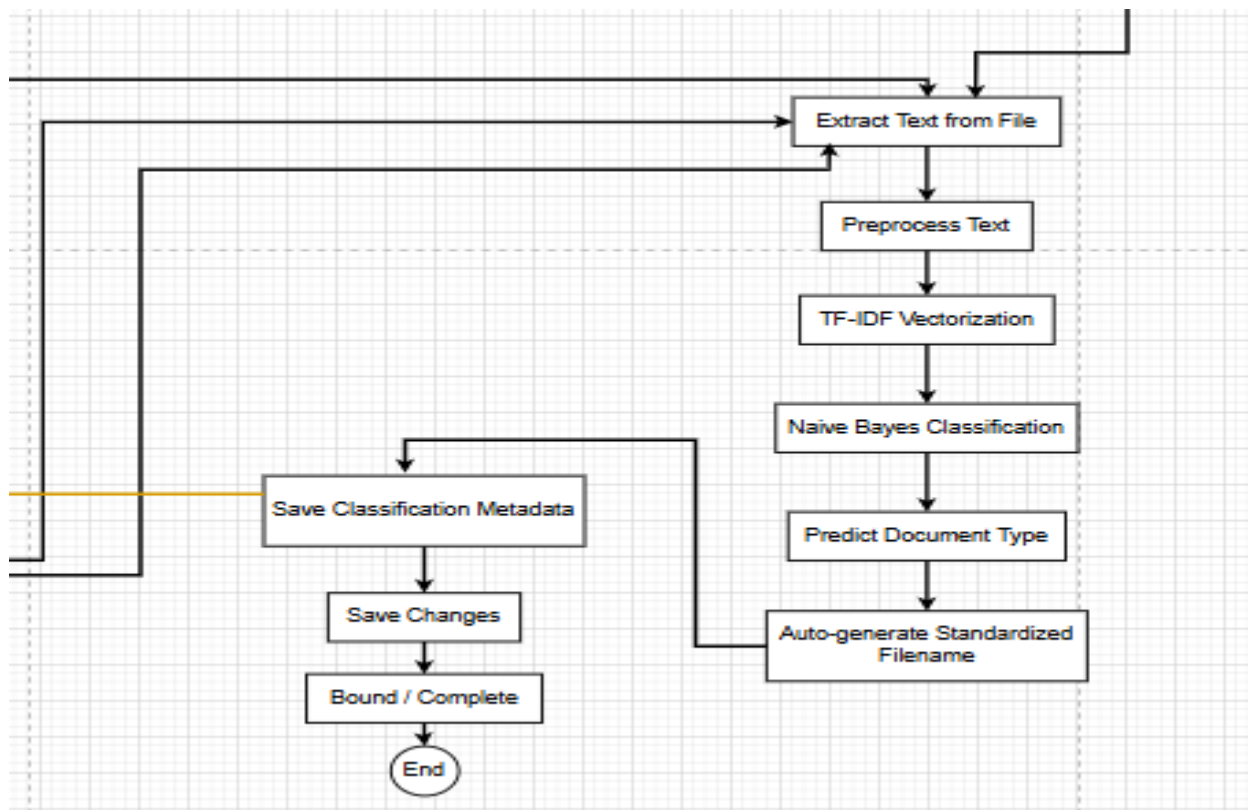
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4. Machine Learning Document Classification Pipeline

- Role: Automated File Handling and Archival Standardization.
- **Function:** This module acts as the intelligent gatekeeper for the system's digital vault. When a deliverable is uploaded, the file bypasses traditional manual sorting and enters an automated Natural Language Processing (NLP) pipeline. The system executes a sequence of operations: Extract Text from File, Preprocess Text, TF-IDF Vectorization, Naive Bayes Classification. By analyzing the textual patterns, the AI accurately predicts the document type (e.g., Proposal, Manuscript, or Journal). It then auto-generates a strictly standardized filename and saves the classification metadata to the database before permanently locking the file into the archive. This guarantees perfect repository organization and ISO audit readiness without requiring human intervention.

Figure 8: ML Flow



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5. Centralized Data Management

- *Function:* All platforms connect to a single, unified database (represented by the yellow connections). This ensures that a grade entered on a mobile phone is instantly reflected on the Admin's dashboard, maintaining data consistency across the department.

Figure 9: Database

